Library Digitization & Book Circulation Management System

A project report

Submitted By

Bhumika Saxena (23BCS12655)

Khyati Chauhan (23BCS13290)

Tanisha (23BCS12582)

In partial fulfillment for the award of the degree of

Bachelor of Engineering (3rd Year)

in

COMPUTER SCIENCE & ENGINEERING

Chandigarh University

May 2026

BONAFIDE CERTIFICATE

Certified that this project report **"LIBRARY DIGITIZATION & BOOK CIRCULATION MANAGEMENT SYSTEM"** is the Bonafide work of **"Bhumika (23BCS12655), Khyati (23BCS13290), and Tanisha (23BCS12582)"** who carried out the project work under my/our supervision.

SIGNATURE OF HOD

Dr.Sanjiv Kumar

HEAD OF THE DEPARTMENT

BE-CSE

SIGNATURE OF THE SUPERVISOR

Er. Rahul Durgapal

SUPERVISOR

Assistant Professor

BE-CSE

Submitted for the project viva-voce examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We take immense pleasure in expressing our sincere gratitude to all those who extended their invaluable support and guidance throughout the completion of this project. The successful execution of Library Digitization & Book Circulation Management System would not have been possible without the encouragement, direction, and cooperation of several individuals.

We express our deepest sense of gratitude to our project supervisor, Er. Rahul Durgapal (E17963), Department of Computer Science and Engineering, Chandigarh University, for his constant guidance, constructive criticism, and relentless encouragement. His insightful suggestions and profound domain knowledge significantly shaped the technical architecture and academic quality of this report.

We are deeply indebted to the Head of the Department of Computer Science and Engineering at Chandigarh University for providing the necessary resources, infrastructure, and an environment conducive to learning and innovation. The laboratory facilities and computing resources made available to us were instrumental in the development and testing of the Library Digitization & Book Circulation Management System.

We also extend our heartfelt thanks to all faculty members of the Department whose teachings in database management systems, web technologies, software engineering, and object-oriented programming formed the bedrock upon which this project was built. Our gratitude extends to our fellow students who participated in testing the application and provided invaluable feedback.

We express our gratitude to our families for their unconditional support, patience, and understanding during the demanding phases of this project. Finally, we acknowledge the open-source community and the developers of technologies such as Java Servlets, JavaServer Pages (JSP), Apache Tomcat, MySQL, JDBC, and Apache Maven, whose freely available tools, libraries, and documentation served as the foundation for building this Library Management System.

Bhumika Saxena| Khyati Chauhan | Tanisha

B.E. Computer Science & Engineering, Chandigarh University – May 2026

ABSTRACT

The fast growth of schools and places to store information has made it more important to have systems that can handle library operations quickly and reliably. Managing a library the old - fashioned way, with paper records, manual registers, and physical catalogs, takes a lot of time, is prone to mistakes, and doesn't work well with large amounts of data. Keeping transaction histories for a long time also takes up a lot of space and makes it hard to find them. To get around these problems, a web-based Library Management System (LMS) has been made to digitize and make library operations more efficient.

Java technologies like JavaServer Pages (JSP), Servlets, JDBC, and MySQL were used to build the proposed system. It runs on the Apache Tomcat server. It uses the Model-View-Controller (MVC) architecture, which keeps the presentation layer, business logic, and data management separate from each other. JSP is used to make user interfaces that change based on what the user does. Servlets handle requests and control flow, and MySQL stores data in structured relational tables that don't go away. Using DAO and Service layers makes the application even more modular, easy to maintain, and able to grow.

The system also uses both client-side and server-side validation methods to make sure that the data is correct and to stop bad inputs. Unit testing, integration testing, and functional testing have all been done on it to make sure it works reliably, consistently, and correctly. The development process uses the Waterfall model of the Software Development Life Cycle (SDLC), which makes it possible to design and implement in a structured and systematic way.

The Library Management System is a complete, flexible, and effective answer for libraries today. It makes things easier for both administrators and students by reducing the amount of work that needs to be done by hand, making data easier to access, and increasing accuracy. The system is also a solid base for future improvements like better management, online reservations, reporting tools, and moving to more advanced frameworks like Spring Boot.

Library Management System (LMS), SDLC, MVC Architecture, Java, JSP, Servlets, JDBC, and MySQL are some of the keywords.

TABLE OF CONTENTS

Chapter	Title	Page No.
1	Introduction	6
1.1	Overview Of Library Management Systems	6
1.2	Problem Statement	6
1.3	Objectives	7
1.4	Scope	7
1.5	Motivation	7
2	Literature Review	9
3	System Analysis	11
4	System Design	13
5	Technology Stack	21
6	Implementation Details	24
7	Module Description	27
8	Working Flow	29
9	Database Design	32
10	Results & Output Screens	35
11	Testing	37
12	Advantages & Limitations	39
13	Future Enhancements	40
14	Conclusion	41
15	References	42

CHAPTER 1: INTRODUCTION

1.1 Overview of Library Management System

This online Library Management System was developed to organize and ease the main functions of a library, including management of books, management of users, and circulation of books. The Library Management System will allow for a central dashboard to view details about the library, such as total number of books, number of books borrowed, number of books returned, and number of users registered so that you can manage library activity efficiently.

It will provide an administrator the ability to complete critical functions such as adding a book, managing users, assigning books to users, and processing returns while maintaining an accurate record of transaction history. The Library Management System will provide you with real-time updates so that you can accurately track the availability of books and the status of your users. The Library Management System also allows multiple users to access the Library Management System at the same time, which will increase the level of efficiency and decrease the amount of manual work.

The Library Management System will also have a secure login for users and will automatically manage new records so that the data is valid and the Library Management System will function smoothly. Overall, the Library Management System provides the easiest, most cost-effective method for library administration.

1.2 Problem Statement

Centralized systems for book, user, and circulation management within libraries are often inefficient because of the lack of a straightforward method for performing these functions. Traditional, cumbersome or complex systems have made it more difficult than necessary to accurately track and process both the issuance and return of library books in real-time, resulting in inaccurate records (as well as errors) and delays. For this reason, there is a need for a cost-effective user-friendly library management system that provides accurate tracking capabilities.

1.3 Objectives of the Project

This project's goal is to have a web-based Library Management System that makes managing books, users, and circulation easier and more automated than ever before. This system will have a single central dashboard to provide the most important information such as the number of books in total, issued books (out) and returned books (in), and the number of registered users. The focus of the system will be to help streamline the process of issuing or returning books while keeping accurate records of book transactions. The project will also provide real-time updates on the availability of books, provide and reduce the amount of manual work involved in managing a library, reduce the number of errors in the library management process, and develop a user-friendly and cost-effective tool for efficiently managing a library.

1.4 Scope of the Project

This project aims to deliver a web-based Library Management System that encompasses all major functions of a library, including the management of books and users, and the circulation of books. The Network Manager will be able to add/edit/delete/maintain books, to register the users, to assign books, to check-in/check-out books, and to maintain all transaction histories.

The central dashboard for the Library Management System will allow the User to monitor various aspects of the library, including the total number of books in the library, the total number of books issued, the total number of books returned, and the total number of registered users.

All books will be updated in real-time and support for multiple users will be available, allowing library personnel to have access to the LMC simultaneously. The Library Management System is also designed to minimize manual labor (massive efficiency improvement) and is primarily targeted at small to medium-sized libraries. Additionally, payment processing, notifications, and role-based access control are not in the current scope of this project; however, they will be considered for future development of the Library Management System.

1.5 Motivation

This project's rationale is to provide a modern, simplified way of managing libraries because traditional methods involve manual processes that take too much time and have many opportunities for mistakes due to their system being divided into registers/book records and other locations. As a

growing number of users require their library to be operated in an organized and dependable way, the need for a system that performs these operations was created.

The main goal of this project is to provide libraries with a user-friendly, cost-effective solution that will help streamline the management of their daily operations. The proposed solution will allow libraries to establish one centralized dashboard for tracking the availability of books in real time, and to make it easier to conduct transactions such as issuing books and returning books, thereby reducing the amount of time, effort, and resources required by libraries to manage their operations. This solution is designed specifically for small and medium-sized libraries that need an uncomplicated yet scalable digital solution.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Systems

Library management systems (LMSs) are computer programs that allow libraries to automate their processes. These systems allow libraries to keep record of their inventory, manage their users and track the items borrowed. The current offerings from many of these systems includes functionality for catalog management, circulation management, reporting and online searches – all of which have improved the efficiency of libraries by improving accessibility to their collections while also reducing the amount of time library staff spend manually entering data.

Many LMSs also have additional functionality such as barcode/RFID integration, automatic report creation and online public access to catalogs (OPAC) which allow users to search for, locate and reserve items digitally. Multiple users can concurrently access the system and all transactions and item availability will update in real-time. Despite these additional features, however, many existing LMSs are complex, expensive and/or require a significant amount of technical expertise for installation and support, thereby making them less well suited for small and mid -sized libraries. Additionally, many existing LMSs do not have a simple user interface and/or a well-defined workflow, which negatively impacts use by non-technical staff in the library. As such, there is a continued need for a successful, cost effective, and easy to use LMS that emphasizes the functionality of the core processes related to the management of the library's collection, users, and items that are within the library.

Platform	Strengths	Limitations
Traditional Manual System	Simple to use, no technical requirement	Time-consuming, error-prone, no real-time tracking
LibSys	Comprehensive features, widely used in institutions	Expensive, not suitable for small libraries.
eGranthalaya	Government-supported, standardized system	Limited customization, complex interface
Proposed System (Project)	Simple UI, dashboard-based, easy book assign/return, cost-effective	Limited advanced features (no role-based, no notifications yet)

2.2 Limitations of Current Systems

Although now more digital than ever, library management systems continue to have several drawbacks. Installation or customization of numerous systems is an intricate process, beyond the technical abilities of most non-technical staff, needing to be maintained by an IT team/individual. Library systems are usually high in cost in terms of setup and maintenance which is a barrier to many small and/or medium-sized libraries.

In addition, many systems do not have user-friendly interfaces or user-friendly dashboards that enable managers to efficiently track activities (e.g., circulation of items, user record management, etc.). Workflows for issuing/returning items are often inefficient, causing delays and inaccurate data, therefore making it difficult for managers to accurately monitor library performances. Limited customization options prevent libraries from being able to adapt systems to the needs of the library. It is necessary to develop a new library management system that is simple, low-cost, and easy to use.

2.3 Need for Library Management System

Many libraries today still use manual systems for managing their collections, or rely on complicated, cumbersome software solutions that are difficult to use and maintain. This can result in poor record keeping, difficulties with tracking circulation, and inefficient daily operations among other problems.

To address this, we require a new integrated solution that is simple to use, cost-effective, and centralized for managing books, users, and transactions. The system will provide real-time tracking of both issued and returned books; a simplified process to assign and return books; and a clear dashboard to view all library activities. By eliminating the amount of manual effort required and increasing accuracy, the library will be able to operate more efficiently and reliably.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Functional Requirements

Functional requirements define the specific behaviours and operations the system must support, identified through stakeholder analysis and domain research.

Req ID	Description
FR-01	System allows user login with username and password.
FR-02	System authenticates users and provides secure access.
FR-03	System displays a dashboard with total books, assigned books, returned books, and users.
FR-04	System allows adding new books to the library.
FR-05	System displays the list of all available books.
FR-06	System allows adding new users.
FR-07	System displays user details and records.
FR-08	System allows assigning books to users.
FR-09	System updates book status when assigned.
FR-10	System allows returning books.
FR-11	System updates records after book return.
FR-12	System shows currently issued books.
FR-13	System maintains all transaction records in the database.
FR-14	System allows multiple users to access the system simultaneously.
FR-15	System provides logout functionality

3.2 Non-Functional Requirements

Attribute	Requirement
Performance	System pages should load within 2–3 seconds under normal usage conditions
Scalability	System should support multiple users accessing the system simultaneously
Security	System provides secure login authentication and protects user data.
Usability	Interface should be simple and easy to use without requiring training
Reliability	System should function consistently without errors during operations.

Maintainability	Code should be modular and easy to update or modify
Compatibility	System should work on major browsers like Chrome, Edge, and Firefox
Data Integrity	System should ensure accurate storage and updating of records in the database

3.2.1 Technical Feasibility

The proposed Library Management System is technically feasible as it is developed using commonly used technologies such as Java (Servlets/JSP), HTML, CSS, JavaScript, and MySQL. These technologies are widely supported, easy to implement, and compatible with standard tools like Eclipse and Apache Tomcat. The system does not require advanced hardware or complex infrastructure. Its simple architecture ensures smooth integration between frontend and backend, making development, deployment, and maintenance practical and achievable.

3.2.2 Economic Feasibility

The system is economically feasible as it is developed using open-source and freely available technologies, which reduces development and implementation costs. No expensive hardware or licensed software is required. The system minimizes manual work, saving time and operational costs for libraries. It is especially suitable for small and medium-sized libraries that need a cost- effective solution for managing their operations efficiently.

3.2.3 Operational Feasibility

The system is operationally feasible as it is designed with a simple and user-friendly interface, making it easy for users to operate without requiring technical expertise or training. Features such as dashboard, book assignment, return process, and user management are straightforward and efficient. The system improves daily library operations by reducing manual effort, ensuring accurate record-keeping, and providing real-time updates, making it highly practical for regular use.

CHAPTER 4: SYSTEM DESIGN

4.1 System Architecture Diagram

The proposed Library Management System follows a three-tier architecture comprising a Presentation Tier (JSP/HTML/CSS frontend), a Business Logic Tier (Java Servlets backend), and a Data Tier (MySQL database). These three tiers communicate through HTTP requests and responses, ensuring a clear separation of concerns and smooth data flow between the user interface, application logic, and database. This architecture improves maintainability, scalability, and overall system performance.

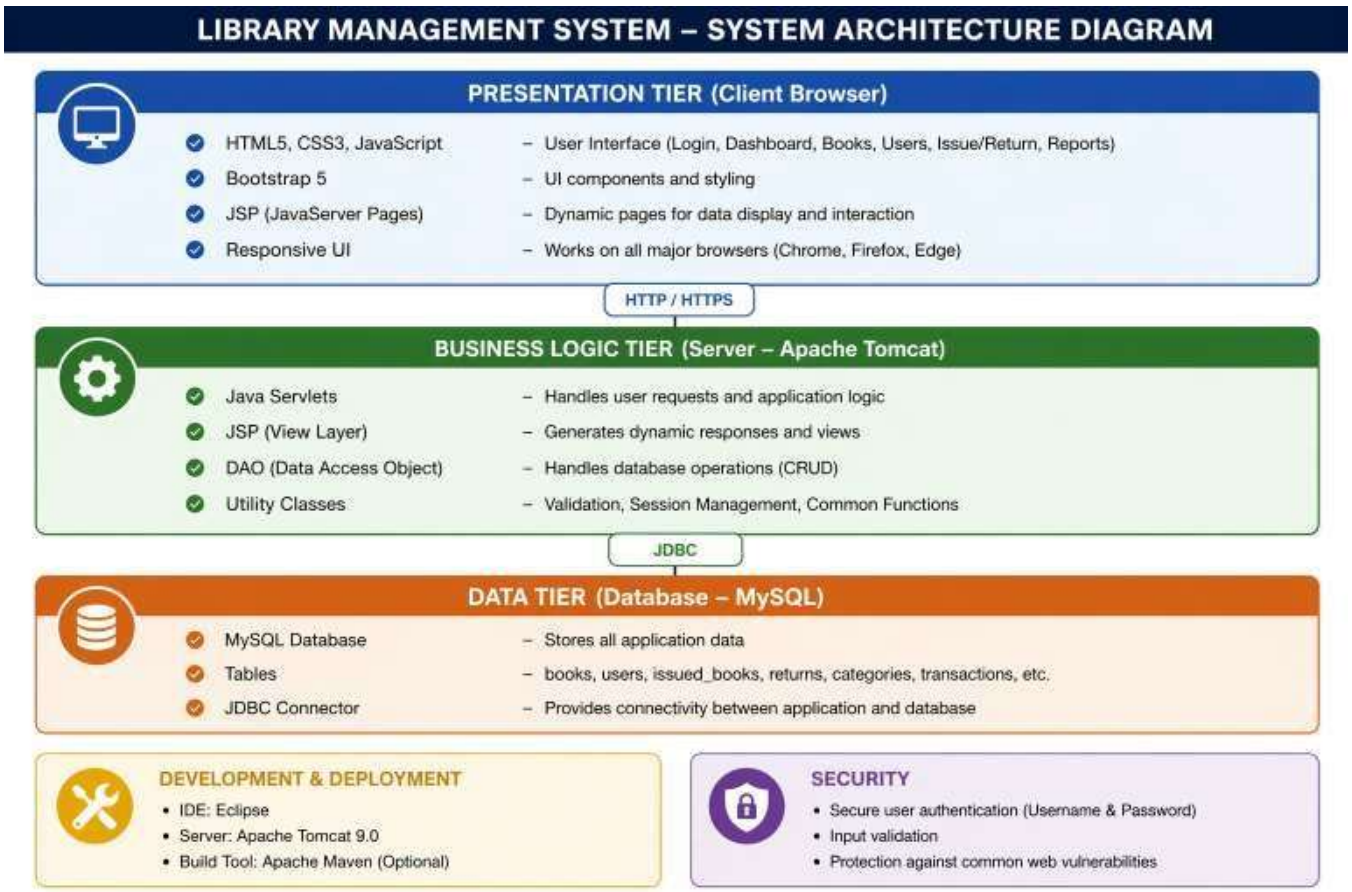

Figure 4.1 – Library Management System Three-Tier System Architecture

The user interface rendering and user interaction operation are managed by the Presentation Tier including, but are not limited to: accessing the login screen, viewing the dashboard, managing books, managing users, and issuing or returning books to users. The Business Logic Tier processes user requests, validates to solve issues prior to executing them, manages application logic, and processes transactions related to issuing or returning books. As it relates to processing user requests and database interactions, the Business Logic Tier uses JDBC to retrieve data from the Data Tier and send data back to be stored within the Data Tier. The Data Tier has been implemented as a relational database to store all book, user, and transaction information in a structured format.
Apache Tomcat is used to deploy the application and the tools & technologies utilized to develop this project include: Eclipse. Overall, the tools and technologies chosen ensure for a streamlined development and deployment process.

4.2 Use Case Diagram

The Use Case Diagram depicts how the Library Administration Management System is interfaced with by the user, who is also the Library System Manager. In the case of this system, all operations that are performed within it are with one or more users (admins) performing those operations through a single interface. There is no role-based access to the Library Administration Management System - i.e., once authenticated, all functions can be performed by a user with proper authorization.

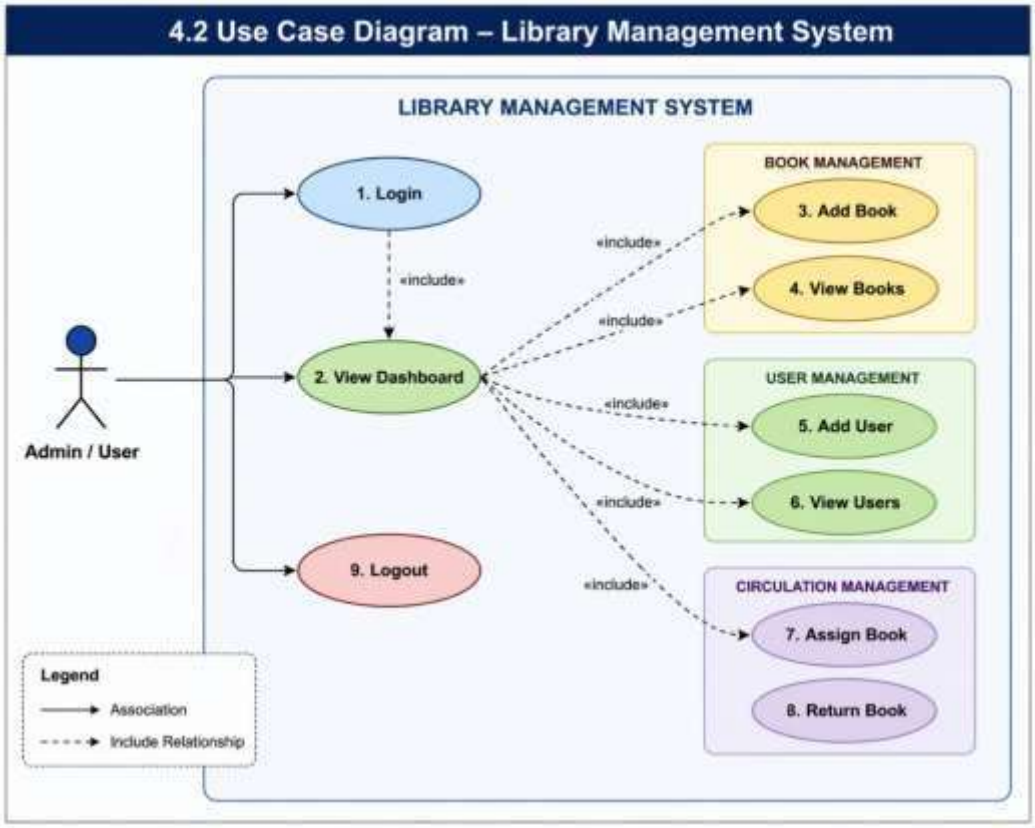

Figure 4.2 – Library Management Use Case Diagram

Once logged into the Library Administration Management System, the Admin/User will be able to access the Library Dashboard, which provides a High-level Overview of the library's activities. The Admin/User will be able to manage Library's Books by adding and viewing books, manage Library's Users by adding and viewing Users, and manage Circulation Operations by Assigning Books and processing Book Returns. The system provides for ease of interactions between users by providing for real time updates to all data in the system so that accurate data is always maintained in the system. At any time, the Admin/User or Library Administrator can securely log out of the Library Administration Management System after completing their Operations.

4.3 Data Flow Diagram – Level 0 (Context Diagram)

The Context Diagram depicts the entire System as one Process and indicates how the System interacts with its external Entities (Admin and Users). The Library Management System is the Process center in this project while Admin and User are the primary External Entities that are

interacting with the Library Management System.

The External Entities (Admin and Users) will provide input such as login credentials, book details, user details and assignment of book as well as return of book information to the Library Management System, which will be processed in the Library Management System to be stored/retrieved from the database. In return, the Library Management System will provide output such as dashboard information, book list, user details, and status of transaction.

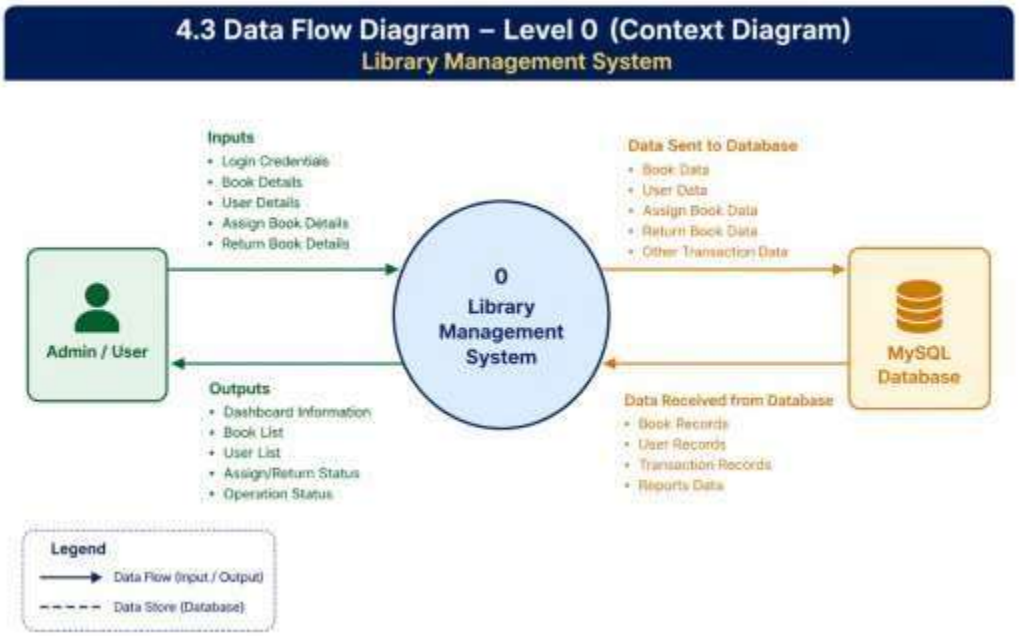

Figure 4.3 – DFD Level 0: Context Diagram

4.4 Data Flow Diagram – Level 1

The Level 1 DFD decomposes the Library Management System into its primary sub -processes, revealing internal data flows and storage interactions. Five processes are identified: Authentication, Dashboard Management, Book Management, User Management, and Book Circulation (Assign/Return), each interacting with the database to manage and update records efficiently.

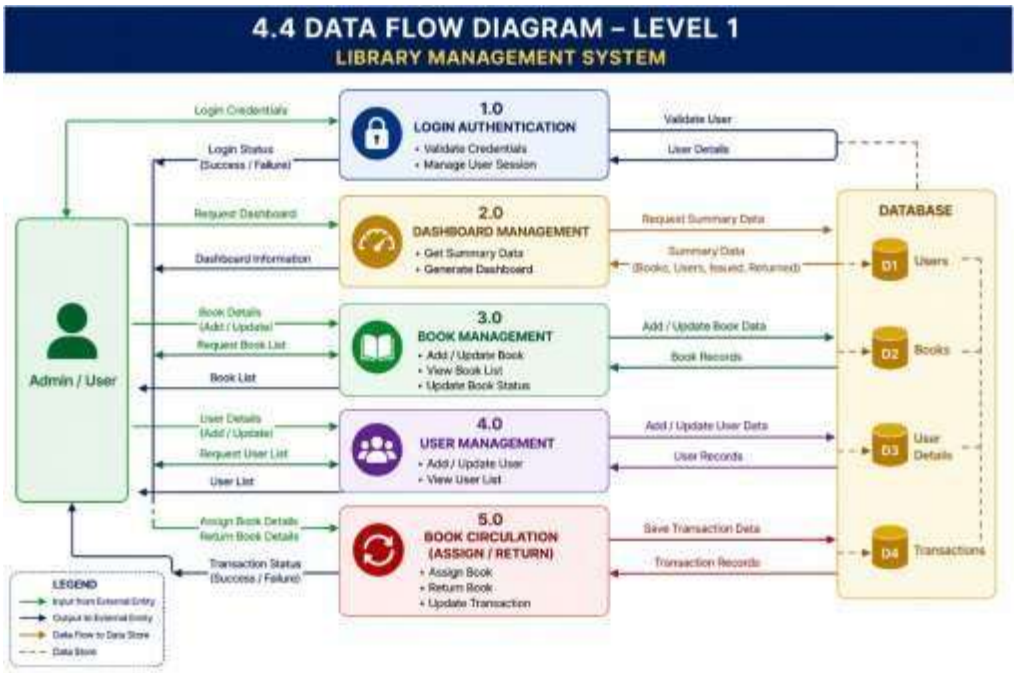

Figure 4.4 – DFD Level 1: Decomposed Processes

4.5 Entity-Relationship (ER) Diagram

The ER Diagram models the data entities in the Library Management System and the relationships between them. Three primary entities are defined — Users, Books, and Book_Issued (Transactions) — each linked through well-defined foreign key relationships that enforce referential integrity. The diagram captures cardinality constraints, where one user and one book can be associated with multiple transaction records, ensuring consistent and accurate data management throughout the system.

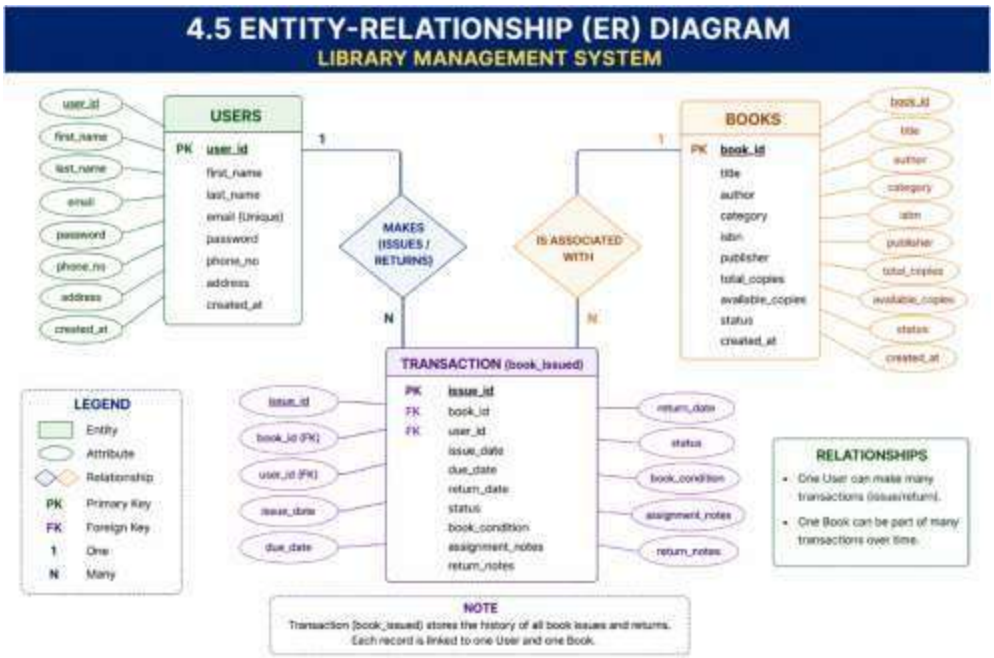

Figure 4.5 – Library Management System Entity-Relationship Diagram

4.6 Sequence Diagram – Book Issue & Return Flow

The sequence diagram shows how the admin/user, jsp based user interface, back-end java servlet (IssuServlet, ReturnServlet and BookServlet) and MySQL database interact when issuing and returning books. When issuing the book, the user enters the information needed to complete the transaction, the system checks if the requested book is available, retrieves the book's details from the database, inserts a row into the book_issued table indicating that the book that has been issued to the user, and updates the number of copies available of that book. When returning the book, the system validates the issued record and updates the details and status of the record being returned to the user, as well as increasing the total number of copies available of that book by one. Each of these transactions are done in sequence to help maintain the integrity of the data as well as the accuracy of the transactions being performed.

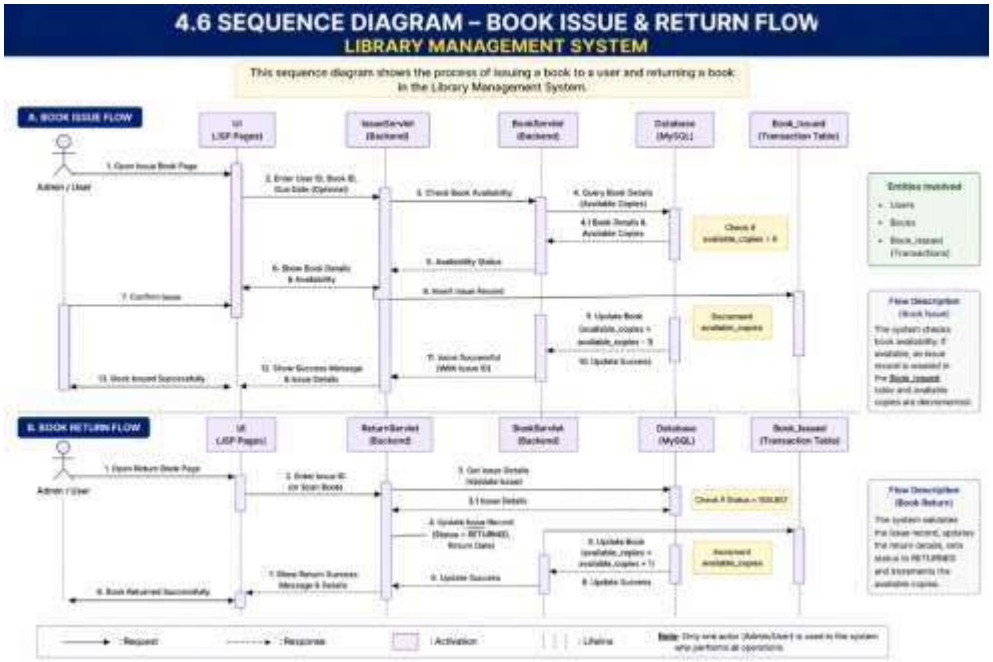

Figure 4.6 – Sequence Diagram: Book issue and return Flow

4.7 Class Diagram

The primary backend classes, their properties, methods and all cross -class relationships are summarized in the Class Diagram. The architecture or layout of this system employs a layered architecture of the DAO (or Data Access Object) pattern where data access logic and business logic are disconnected from each other. The Service layer is used to perform all business operations as well as to interact with DAO classes; while the Controller classes perform the actual user request/response. This way of designing provides for an organized, manageable and efficient use of data within the system.

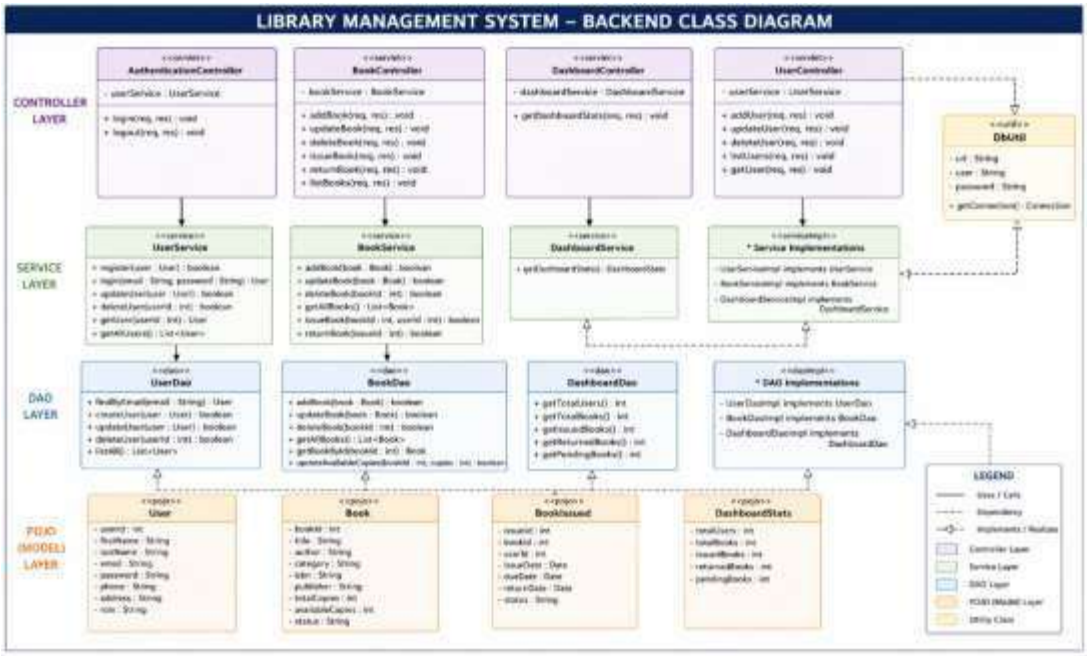

Figure 4.7 – Backend Class Diagram

CHAPTER 5: TECHNOLOGY STACK

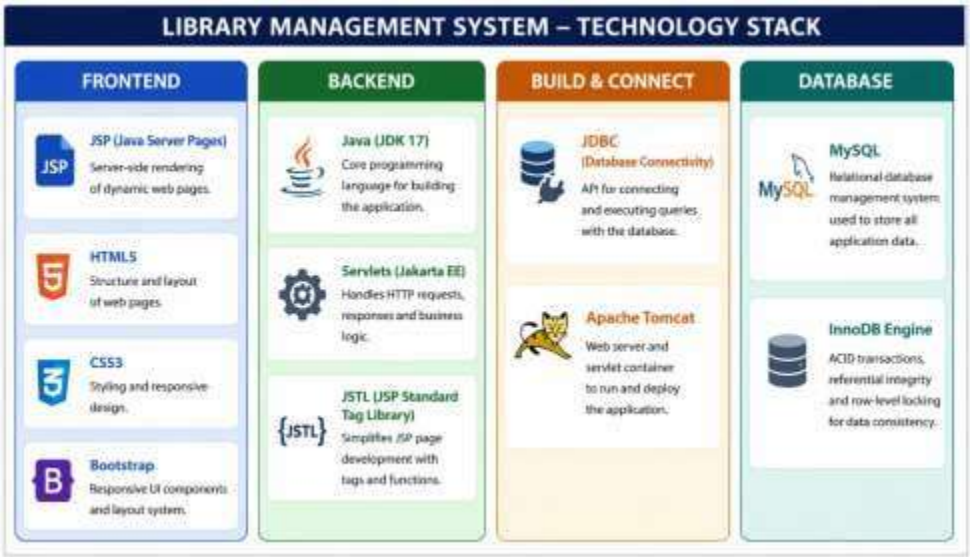

Figure 5.1 – Library Management System - Technology Stack

5.1 JSP (JavaServer Pages)

JSP (JavaServer Pages) is a server-side technology used for creating dynamic web pages in Java-based applications. It allows embedding Java code within HTML to generate dynamic content at runtime. JSP simplifies the presentation layer by separating UI design from business logic. In the Library Management System, JSP is used to build interfaces such as login, dashboard, book management, and user management pages. It works together with Servlets, where JSP handles the view layer and displays data received from backend processing.

5.2 HTML5

HTML5 is the standard markup language used to structure web pages and define their content. It provides elements for forms, tables, navigation, and multimedia integration. In this system, HTML5 is used to design the basic structure of all web pages, including forms for adding users, managing books, and issuing/returning books. It ensures proper layout and accessibility across different browsers.

5.3 CSS3

CSS3 is used for styling and enhancing the visual appearance of web pages. It enables the use of colors, layouts, animations, and responsive design techniques. In the Library Management System, CSS3 is used to design the dashboard, navigation sidebar, and forms, ensuring a clean and user-friendly interface.

5.4 Bootstrap

Bootstrap is a front-end framework that provides pre-designed components and responsive layouts. It simplifies UI development by offering ready-to-use classes for buttons, forms, grids, and navigation elements. In this project, Bootstrap is used to create a responsive and consistent user interface that works efficiently across different screen sizes.

5.5 Java (JDK 17)

Java is the core programming language used to develop the backend of the system. It provides object-oriented features, platform independence, and robust performance. In this project, Java is used to implement business logic, manage application flow, and integrate various components of the system.

5.6 Java Servlets

Servlets are server-side components that handle HTTP requests and responses. They act as the controller in the MVC architecture, processing user inputs and interacting with the service and DAO layers. In the Library Management System, Servlets manage operations such as user authentication, book management, and issue/return workflows.

5.7 JSTL (JSP Standard Tag Library)

JSTL is a collection of standard tags used in JSP to simplify the development of dynamic web pages. It reduces the need for Java code inside JSP pages by providing tags for iteration, condition handling, and formatting. In this system, JSTL is used to display dynamic data such as book lists and user records efficiently.

5.8 JDBC (Java Database Connectivity)

JDBC is an API used to connect Java applications with relational databases. It enables execution of SQL queries and management of database operations. In this project, JDBC is used within DAO classes to perform CRUD operations on users, books, and transaction records, ensuring secure and efficient data handling.

5.9 Apache Tomcat

Apache Tomcat is a web server and servlet container used to deploy and run Java web applications. It processes HTTP requests and executes JSP and Servlet components. In this system, Tomcat is used to host the application and manage communication between the client and server.

5.10 MySQL

MySQL is a relational database management system used to store and manage application data. It organizes data into structured tables and supports SQL queries for data manipulation. In this project, MySQL stores information related to users, books, and book issue/return transactions.

5.11 InnoDB Storage Engine

InnoDB is the default storage engine of MySQL that provides ACID compliance and ensures data integrity. It supports transactions, foreign keys, and row-level locking. In this system, InnoDB ensures reliable and consistent database operations during book issue and return processes.

CHAPTER 6: IMPLEMENTATION DETAILS

6.1 Frontend Structure

The frontend of the Library Management System is structured using multiple HTML/JSP pages organised according to a feature-based directory structure. The primary entry point is the *login page*, which allows users to authenticate and access the system. Each page corresponds to a specific functionality such as dashboard, book management, user management, and book issue/return operations. Navigation between pages is handled through server -side routing using Servlets, where each request is processed and the appropriate page is rendered. Shared styling is maintained through centralized CSS files, ensuring a consistent and responsive user interface. This approach provides a clear separation of concerns and simplifies navigation and maintenance of the application.

File/Package	Purpose
assets/css/auth.css	Provides styling for authentication pages like login and register
assets/css/common.css	Contains common styles used across dashboard and all modules
login.html	Allows user to log into the system.
register.html	Enables new user registration.
dashboard.html	Displays system overview such as total books, users, and transactions
books.html	Shows list of all books in the system.
add_book.html	Provides form to add new books.
edit_book.html	Allows editing existing book details
assign_book.html	Used to assign (issue) books to users
users.html	Displays list of all users
add_user.html	Form to add new users
edit_user.html	Allows updating user information.
return_book.html	Handles book return process
show_return_book_detail.html	Displays details of returned books
error.html	Displays error messages and handles invalid requests

6.2 Backend Structure

The backend of the Library Management System is structured using a layered architecture following the MVC design pattern. The system is organised into packages such as controller, service, dao, pojo, filter, and util, each responsible for a specific functionality. The controller layer handles user requests through classes like AuthenticationController, BookController, UserController, and DashboardController, coordinating application flow. The service layer contains business logic, while the DAO layer manages database interactions using JDBC. The pojo classes represent data entities such as User, Book, and BookIssued, enabling smooth data transfer across layers. Additionally, utility classes handle database connectivity, and filters manage authentication and access control. This modular design ensures clear separation of concerns, scalability, and maintainability of the system.

File/Package	Purpose
controller/*	Handles all user requests (authentication, book, user, dashboard operations)
service/*	Contains business logic for books, users, and dashboard
serviceImpl/*	Implements service layer functionality
dao/*	Performs database operations using JDBC
daoImpl/*	Implements DAO layer logic
pojo/*	Represents entities such as User, Book, and BookIssued
filter/AuthFilter.java	Handles authentication and access control
util/DbUtil.java	Manages database connection

6.3 Servlet Endpoints

The system uses Servlet-based endpoints defined through the @WebServlet annotation to handle client requests. Each controller processes requests using HTTP methods such as GET and POST, and different operations are managed using an action parameter. Instead of RESTful APIs, the application follows a request-response model where Servlets interact with the service and DAO layers to perform required operations. Authentication and access control are managed using session tracking to ensure secure access to the system.

Endpoint	Method	Description	Access
/AuthenticationController?action=login	POST	Authenticates user and starts session	Public
/AuthenticationController?action=logout	GET	Logs out user and invalidates session	Authenticated
/UserController?action=list	GET	Displays list of users	Authenticated
/UserController?action=add	POST	Adds a new user	Authenticated
/UserController?action=edit	POST	Updates user details	Authenticated
/BookController?action=list	GET	Displays list of books	Authenticated
/BookController?action=add	POST	Adds a new book	Authenticated
/BookController?action=edit	POST	Updates book details	Authenticated
/BookController?action=assign	POST	Issues book to user	Authenticated
/BookController?action=return	POST	Handles book return process	Authenticated
/DashboardController	GET	Displays dashboard statistics	Authenticated

CHAPTER 7: MODULE DESCRIPTION

7.1 Authentication Module

The Authentication Module serves as the entry point of the Library Management System, managing user login and session control. When a user submits login credentials, the Authentication Controller validates them against the database and establishes a session upon successful verification. The AUTH Filter ensures that only authenticated users can access protected resources by intercepting incoming requests and checking session validity before allowing further processing.

If authentication fails, the system redirects the user back to the login page with appropriate error messages. Upon logout, the session is invalidated to prevent unauthorized reuse. This module ensures controlled access to the system and maintains security by restricting operations only to verified users.

7.2 Book Issue and Return Module

The Book Issue and Return Module acts as the transactional core of the Library Management System, managing the complete lifecycle of book circulation. When a user issues a book, the Book Controller verifies availability, retrieves book details, and creates a new record in the *book_issued* table while simultaneously decreasing the available copies of the book. This ensures that book allocation is accurately reflected in real time.

During the return process, the system validates the issued record and updates the return date and status accordingly. The available copies of the book are incremented, restoring its availability for future users. All operations are executed sequentially through the service and DAO layers to maintain data consistency and ensure that no partial updates affect the integrity of the system.

7.3 User Management Module

The User Management Module is responsible for handling all user-related operations within the system. It allows adding new users, updating existing user details, and retrieving user information through the UserController. The module interacts with the database via the service and DAO layers to ensure accurate storage and retrieval of user records.

Each operation is validated before execution to prevent invalid or duplicate data entries. The system ensures that user information is consistently maintained and readily accessible for book assignment and tracking purposes. This module plays a key role in managing user data efficiently within the application.

7.4 Book Management Module

The Book Management Module handles all operations related to maintaining book inventory in the system. It allows adding new books, updating existing records, and retrieving book details through the Book Controller. The module ensures that attributes such as title, author, and available copies are accurately stored and maintained.

Whenever changes are made to book records, the system updates the database through the DAO layer, ensuring consistency across all modules. This module plays a crucial role in maintaining an up-to-date and reliable inventory, which is essential for smooth book circulation and tracking.

CHAPTER 8: WORKING FLOW

8.1 User Login and Session Management

The login flow begins when the user enters their registered email and password on the login page. The frontend submits a POST request to the Authentication Controller with the entered credentials. The controller retrieves the corresponding user record from the database and validates the credentials against the stored values. If the credentials are valid, a session is created and the user is redirected to the dashboard.

Once authenticated, the system maintains the user session to allow access to protected resources. The AuthFilter intercepts incoming requests and verifies whether a valid session exists before granting access to any module. If the session is invalid or expired, the user is redirected back to the login page, ensuring secure access control throughout the application.

8.2 Book Issue Process Flow

The booking process follows a structured sequential workflow designed to minimize user error and ensure transactional integrity. The process begins with the user browsing the event catalogue and selecting an event of interest. After viewing the event's interactive seat map and selecting desired seats, the frontend dispatches a seat reservation request to /seat/reserve, which updates seat statuses and initiates the twenty-four-hour payment countdown. The user is then redirected to the Payment Page to review the reservation details and complete payment. Upon payment confirmation, the system executes the booking transaction atomically and redirects the user to the booking confirmation page with a booking reference number.

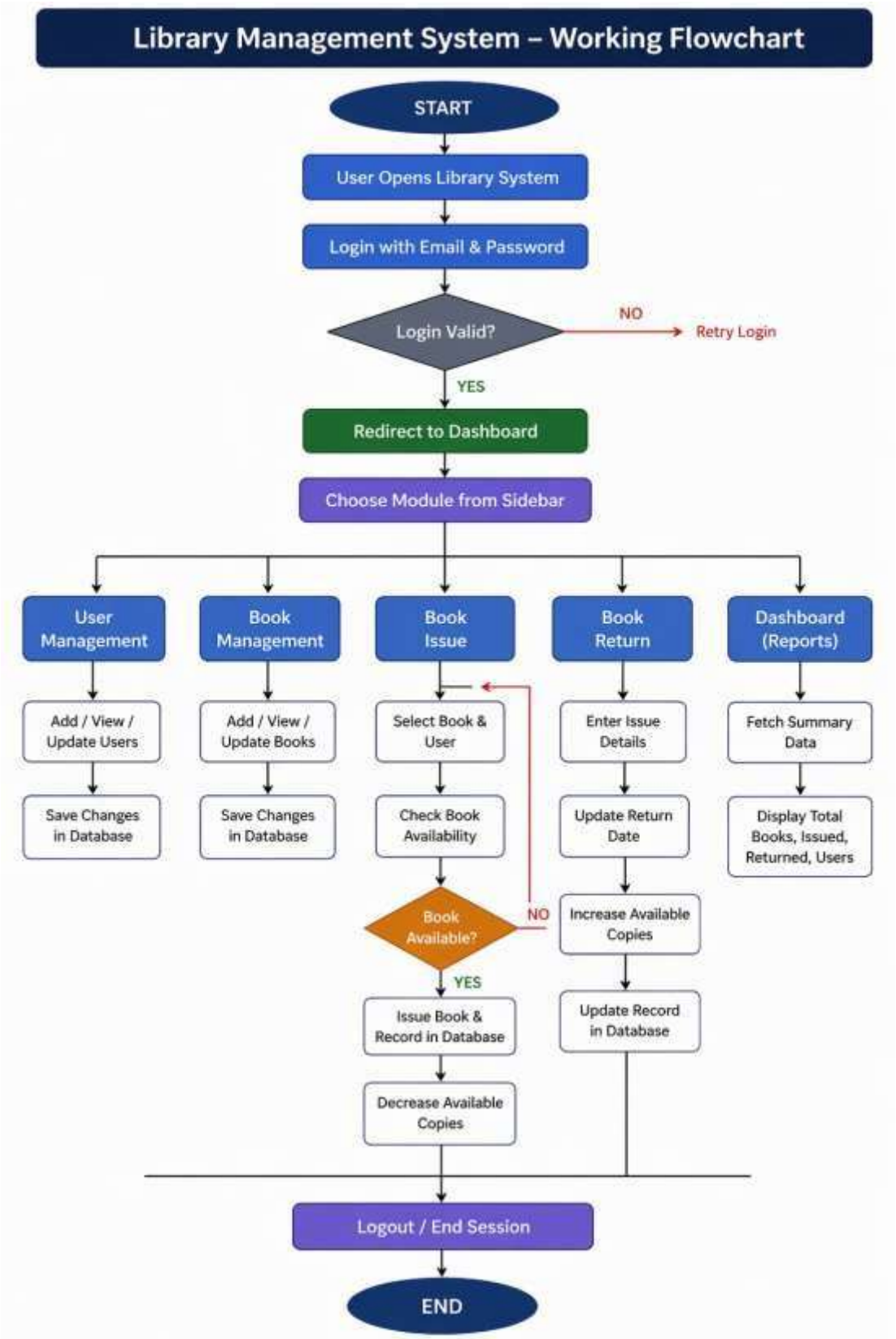

Figure 8.1 – Process Flowchart

8.3 Book Return Flow

The book return process ensures proper tracking and updating of issued books. When a user initiates a return request, the system retrieves the corresponding issue record using the provided details. The request is processed by the Book Controller, which validates the transaction before proceeding.

Once validated, the system updates the return date and status in the *book_issued* table and increments the available copies of the book. This ensures that the book becomes available for future use. The operation maintains data consistency and reflects updates immediately across the system.

8.4 Dashboard and Data Flow

The dashboard flow is designed to provide a real-time overview of system activities. When the user accesses the dashboard, a request is sent to the Dashboard Controller, which retrieves aggregated data such as total books, issued books, returned books, and total users from the database.

The retrieved data is processed through the service and DAO layers and displayed on the frontend. This flow ensures that users can quickly monitor system status and make informed decisions. The dashboard is updated dynamically based on the latest transactions and database records.

CHAPTER 9: DATABASE DESIGN

9.1 Database Schema Overview

The Library Management System database is implemented in MySQL and comprises three primary tables designed using normalization principles to minimize data redundancy and ensure efficient data organization. These tables maintain structured relationships to support core functionalities such as user management, book inventory, and book circulation operations. The schema is designed to provide consistency and scalability while handling routine library transactions.

Foreign key constraints are used to maintain referential integrity between related tables, particularly linking users and books through the book_issued table. The InnoDB storage engine is utilized to support ACID properties and ensure reliable transaction handling. This enables accurate tracking of book issue and return processes while maintaining data consistency across concurrent operations.

9.2 Key Table Definitions

9.2.1 Users Table

The users table stores all registered user information, including personal and authentication details. Each user is uniquely identified by a user_id , which is auto-generated. The table also maintains fields such as name, email, password, role, and contact details.

The email field is defined as unique to prevent duplicate registrations, while the role attribute distinguishes between different types of users (e.g., ADMIN and USER). The created_at field records the timestamp of user registration, enabling tracking of user activity over time.

9.2.2 Books Table

The books table maintains all information related to books available in the library. Each book is uniquely identified by book_id and includes details such as title, author, category, ISBN, and publisher.

The table also stores total_copies and available_copies to track book inventory. The status field indicates the availability of the book, while the created_at timestamp records when the book was added to the system. This design ensures accurate inventory management and supports efficient book tracking.

9.2.3 Book IssuedTable

The book_issued table records all book issue and return transactions, acting as a bridge between users and books. Each transaction is uniquely identified by issue_id and includes references to both user_id and book_id through foreign key constraints.

This table stores details such as issue date, due date, return date, and status of the transaction. Additional fields like book_condition, assignment_notes, and return_notes provide flexibility for recording extra information. The foreign key relationships ensure referential integrity and maintain consistency between related tables.

Table	Primary Key	Foreign Keys	Description
Users	user_id (INT, AI)	_____	Stores user details including name, email, password, role, and contact information
Books	book_id (INT, AI)	_____	Contains book information such as title, author, category, and availability
Book_issued	issue_id (INT, AI)	user_id → users, book_id → books	Stores book issue and return transactions linked to users and books

9.3 Sample Data Records

1. Users Table

user_id	first_name	last_name	email	role	phone_no	address	created_at
1	Admin	Test	admin@library.com	ADMIN	84848454545	Delhi	2026-04-01 10:00:00
2	Rahul	Sharma	rahul@gmail.com	USER	9876543210	Mumbai	2026-04-02 11:30:00

2. Books Table

book_id	title	author	category	isbn	publisher	total_copies	available_copies	status	created_at
1	Java Programming	James Gosling	Programming	123456789	Oracle	10	8	AVAILABLE	2026-04-01 09:00:00
2	Data Structures	Mark Allen	CS	987654321	Pearson	5	5	AVAILABLE	2026-04-02 10:15:00

3. Book Issued Table

issue_id	book_id	user_id	issue_date	due_date	return_date	status	book_condition	assignment_notes	return_notes
1	1	2	2026-04-05 12:00:00	2026-04-12 12:00:00	NULL	ISSUED	Good	First issue	NULL
2	2	2	2026-04-01 10:00:00	2026-04-08 10:00:00	2026-04-07 09:00:00	RETURNED	Good	—	Returned on time

CHAPTER 10: RESULTS AND OUTPUT SCREENS

This chapter describes the user interface screens and system behavior across all major functional areas of the Library Management System. Each screen is designed using HTML, CSS, JSP, and Bootstrap to ensure a clean layout, responsiveness, and ease of use. The interface follows a dashboard-based structure with consistent navigation and styling across all modules.

10.1 Login Screen

The Login Page presents a centred form layout with fields for email and password, along with a login button styled using Bootstrap components. Input validation ensures that users enter valid credentials, and incorrect login attempts display appropriate error messages on the screen. Upon successful authentication, the system creates a session for the user and redirects them to the dashboard. The login process ensures that only authorized users can access the system, and protected pages are secured using session-based authentication mechanisms.

10.2 User Dashboard and Event Browser

The Dashboard provides a summary view of the system, displaying key metrics such as total books, issued books, returned books, and total users. These values are presented in a structured card layout, allowing users to quickly understand system status. The dashboard also serves as the main navigation hub, providing access to different modules through a sidebar. This improves usability by allowing users to move seamlessly between book management, user management, and transaction operations.

10.3 Book Management Screen

The Book Management interface displays a list of all books in a tabular format, including details such as title, author, category, and availability status. Users can add new books, edit existing records, and view complete book details. The form for adding or editing books includes fields for book information such as title, author, ISBN, publisher, and number of copies. Upon submission, the system validates the input and updates the database accordingly.

10.4 User Management Screen

The User Management screen displays all registered users in a structured table format. It includes user details such as name, email, phone number, and role, allowing administrators to manage users efficiently. The interface provides options to add new users and update existing user details. These operations are handled through forms with proper validation, ensuring accurate data entry and maintenance.

10.5 Book Issue Screen

The Book Issue screen allows users to assign books to registered users. It provides input fields for selecting a user and a book, along with optional details such as due date. The system checks book availability before issuing. If the book is available, the transaction is recorded, and the available copies are reduced. A success message is displayed upon completion of the operation.

10.6 Book Return Screen

The Book Return screen enables users to return issued books by entering issue details or selecting from issued records. The system retrieves the corresponding transaction and validates it before processing. Once the return is confirmed, the system updates the return date, changes the status to returned, and increases the available copies of the book. The updated data is reflected immediately in the system.

CHAPTER 11: TESTING

11.1 Unit Testing

Unit testing in the Library Management System focuses on verifying the correctness of individual components such as DAO and Service layer classes. Each DAO method is tested to ensure that SQL queries are executed correctly and data is properly inserted, updated, and retrieved from the MySQL database. The service layer is tested to validate business logic such as book availability checks, issue and return operations, and dashboard data calculations.

Since the project is based on Servlets and JSP, testing is primarily performed manually and through controlled inputs. Individual methods in controllers are verified by simulating requests and checking expected outputs. This ensures that each component functions correctly in isolation before integration.

11.2 Integration Testing

Integration testing verifies the interaction between different components of the system, including controllers, services, DAOs, and the database. When a request is made from the frontend (JSP page), it is processed by the controller, passed to the service layer, and then to the DAO layer for database operations.

The results are then returned back through the same flow and displayed on the user interface. This process ensures that all layers of the application work together correctly. Integration testing confirms that data flows properly between modules and that system behaviour matches expected outcomes.

11.3 Test Cases

TC ID	Test Case Description	Expected Result	Status
TC-01	User login with valid credentials	Login successful; redirect to dashboard	PASS
TC-02	User login with invalid credentials	Error message displayed	PASS
TC-03	Access dashboard without login	Redirect to login page	PASS
TC-04	Add new user with valid details	User added successfully	PASS

TC-05	Update user details	User information updated	PASS
TC-06	View user list	List of users displayed correctly	PASS
TC-07	Add new book with valid details	Book added successfully	PASS
TC-08	Update book details	Book information updated	PASS
TC-09	View book list	List of books displayed	PASS
TC-10	Issue book with available copies	Book issued; available copies reduced	PASS
TC-11	Issue book when not available	Error message displayed	PASS
TC-12	Return issued book	Book returned; available copies increased	PASS
TC-13	View dashboard statistics	Correct data displayed	PASS
TC-14	Unauthorized access to protected page	Access denied / redirect to login	PASS

CHAPTER 12: ADVANTAGES AND LIMITATIONS

12.1 Advantages

The Library Management System offers a centralized and efficient way to manage library operations, reducing manual work and minimizing errors. It automates key processes such as user management, book inventory tracking, and book issue/return transactions, ensuring faster and more accurate record handling. The use of a database ensures data consistency and real-time updates across the system.

The system is designed with a user-friendly interface, making it easy to navigate and operate without extensive training. Its modular architecture (Controller, Service, DAO) improves maintainability and scalability, allowing future enhancements. Additionally, session -based authentication ensures secure access, protecting sensitive data from unauthorized users.

12.2 Limitations

Despite its benefits, the system has certain limitations. It does not support advanced features such as online reservations, fine calculation for late returns, or notification systems. The absence of role -based access control limits flexibility in managing different types of users with varying permissions.

The system is dependent on a local server environment and does not support cloud deployment or mobile access. Additionally, manual testing is primarily used instead of automated testing frameworks, which may limit testing coverage. Future improvements can address these limitations to make the system more robust and feature-rich.

CHAPTER 13: FUTURE ENHANCEMENTS

13.1 Fine Management System

Introducing an automated fine calculation system for overdue books would enhance accountability and improve book return compliance. The system can calculate fines based on the number of days past the due date and predefined penalty rules. This would require extending the *book_issued* table with fine-related attributes and implementing backend logic to compute and update fines dynamically. The enhancement would also include displaying fine details on the user interface and allowing administrators to manage or waive penalties when necessary. This feature would significantly improve circulation discipline and ensure timely returns of books.

13.2 Online Book Reservation System

Implementing an online reservation system would allow users to reserve books that are currently unavailable. When a book is returned, the system can notify the next user in the reservation queue. This would require maintaining a reservation queue and updating book availability logic. This enhancement would improve user convenience and ensure fair allocation of high-demand books. It would also reduce manual effort in managing waiting lists and enhance overall system usability.

13.3 Email and Notification Integration

Integrating email or SMS notifications would enable the system to automatically inform users about important events such as book issue confirmation, due date reminders, and return acknowledgments. This can be implemented using email APIs or messaging services. Automated reminders for due dates would help reduce overdue cases and improve user engagement. This feature would make the system more interactive and user-friendly.

13.4 Barcode/QR Code Integration

Introducing barcode or QR code scanning for books would automate the issue and return process. Each book can be assigned a unique code that can be scanned using a barcode scanner or mobile device, reducing manual data entry. This enhancement would improve efficiency, reduce errors, and speed up transactions, especially in large libraries with high book circulation.

CHAPTER 14: CONCLUSION

The Library Management System developed in this project provides an efficient and structured solution for managing essential library operations. It automates key processes such as user management, book inventory tracking, and book issue and return, reducing manual effort and minimizing errors. By digitizing these activities, the system ensures accurate record keeping and improves the overall efficiency of library operations.

The system is built using Java web technologies including JSP, Servlets, JDBC, and MySQL, following a layered architecture that separates presentation, business logic, and data access components. This design enhances maintainability, scalability, and modularity, allowing the system to be easily extended in the future. The use of a relational database ensures data consistency and supports real-time updates across all modules.

The user interface is designed to be simple, intuitive, and responsive, enabling users to navigate through different modules such as dashboard, book management, user management, and book circulation with ease. The system also incorporates session-based authentication to ensure secure access, restricting unauthorized users from accessing protected resources.

Although the system successfully implements all core functionalities, there is scope for further enhancement. Features such as fine calculation for overdue books, notification systems, barcode integration, and cloud deployment can be added to improve functionality and scalability. Overall, the project demonstrates a practical implementation of a web-based library management system and highlights the effectiveness of using Java technologies for real-world applications.

CHAPTER 15: REFERENCES

[1] R. T. Fielding, *Architectural Styles and the Design of Network-based Software Architectures*, Ph.D. Dissertation, University of California, Irvine, 2000.

[2] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*. Reading, MA: Addison-Wesley, 1994.

[3] Oracle Corporation, *Java SE Documentation*, 2024. [Online]. Available: https://docs.oracle.com/javase/. [Accessed: Apr. 10, 2026].

[4] Apache Software Foundation, *Apache Tomcat Documentation*, 2024. [Online]. Available: https://tomcat.apache.org. [Accessed: Apr. 10, 2026].

[5] Oracle Corporation, *MySQL 8.0 Reference Manual*, 2024. [Online]. Available: https://dev.mysql.com/doc/. [Accessed: Apr. 10, 2026].

[6] Jakarta EE, *Servlet and JSP Documentation*, 2024. [Online]. Available: https://jakarta.ee. [Accessed: Apr. 10, 2026].

[7] Bootstrap, *Bootstrap Documentation*, 2024. [Online]. Available: https://getbootstrap.com. [Accessed: Apr. 10, 2026].

[8] Eclipse IDE, *Eclipse IDE Documentation*, 2024. [Online]. Available: https://www.eclipse.org. [Accessed: Apr. 10, 2026].

[9] MySQL Workbench, *MySQL Workbench Documentation*, 2024. [Online]. Available: https://www.mysql.com/products/workbench/. [Accessed: Apr. 10, 2026].

[10] W3C, *HTML5 Specification*, 2024. [Online]. Available: https://www.w3.org. [Accessed: Apr. 10, 2026].

[11] W3C, *CSS3 Specification*, 2024. [Online]. Available: https://www.w3.org. [Accessed: Apr. 10, 2026].

[12] I. Sommerville, *Software Engineering*, 10th ed. Harlow, UK: Pearson Education, 2016.

[13] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach* , 9th ed. New York: McGraw-Hill Education, 2020.

[14] C. Coronel and S. Morris, *Database Systems: Design, Implementation, & Management*, 13th ed. Boston: Cengage Learning, 2019.

[15] JDBC, *JDBC API Documentation*, 2024.

[16] GitHub, *GitHub Documentation*, 2024. [Online]. Available: https://github.com.